

1. A 42-year-old man who is HIV positive comes to the physician for immunizations prior to a trip to Mexico. He is leaving in 6 weeks during autumn. He plans to engage in typical tourist activities in metropolitan areas. He takes highly active antiretroviral therapy. Three years ago, he had pneumococcal pneumonia and received the pneumococcal polysaccharide vaccine, 23-valent, at that time. He received a tetanus booster 5 years ago and the hepatitis A and B vaccines 3 years ago. He had all appropriate childhood vaccinations. His vital signs are within normal limits. Examination shows no abnormalities. Laboratory studies show a CD4+ T-lymphocyte count of $643/\text{mm}^3$ (Normal ≥ 500) and an undetectable plasma HIV viral load. Which of the following vaccinations is most appropriate at this time?

- ☐ A) Human papillomavirus
- ☐ B) Inactivated influenza virus
- ☐ C) Measles-mumps-rubella virus
- ☐ D) Meningococcal
- ☐ E) Rabies
- ☐ F) No vaccines are indicated at this time

2. A 92-year-old woman is brought to the emergency department 30 minutes after she was found confused in her home. The weather has been extremely hot, and the temperature in her room was 97°F. She has multi-infarct (vascular) dementia and had a myocardial infarction 30 years ago. She is stuporous. Her temperature is 39°C (102.2°F), pulse is 130/min and regular, respirations are 18/min, and blood pressure is 80/60 mm Hg. Examination shows dry mucous membranes. There is no jugular venous distention. The lungs are clear on auscultation. A grade 2/6 holosystolic murmur is heard at the lower left sternal border. The remainder of the examination shows no abnormalities. Test of the stool for occult blood is negative. Which of the following age-related changes is the most likely explanation for the findings in this patient?

- ☐ A) Decreased airway elasticity
- ☐ B) Decreased baroreflex
- ☐ C) Decreased mucociliary clearance
- ☐ D) Decreased thyroid function
- ☐ E) Diastolic cardiac dysfunction
- ☐ F) Impaired cardiac response to exercise
- ☐ G) Impaired T-lymphocyte function
- ☐ H) Impaired thirst
- ☐ I) Increased lung compliance
- ☐ J) Renal salt wasting

3. A 32-year-old woman is brought to the emergency department because of fever and confusion for 4 hours. She has a 5-year history of systemic lupus erythematosus. She had *Neisseria meningitidis* meningitis 3 years ago. Current medications include hydroxychloroquine. Her temperature is 40°C (104°F). Examination shows nuchal rigidity. Kernig sign is present. Cerebrospinal fluid analysis shows cloudiness. A Gram stain shows gram-negative diplococci. Which of the following mechanisms is the most likely cause of these findings?

- ☐ A) Common variable immunodeficiency
- ☐ B) Deficiency of terminal components of complement
- ☐ C) Impaired cell-mediated immunity
- ☐ D) Leukocyte adhesion deficiency
- ☐ E) Phagocytic cell oxidative metabolism deficiency

4. A previously healthy 25-year-old man comes to the physician because of a 10-day history of fever, sore throat, and generalized fatigue. He takes no medications. He became sexually active with a new female partner 6 weeks ago and does not use condoms. He does not use intravenous drugs. His temperature is 37.8°C (100°F), pulse is 84/min, respirations are 14/min, and blood pressure is 126/76 mm Hg. There are numerous erythematous macules over the upper chest. Examination shows multiple, less than 1-cm, cervical, occipital, and axillary lymphadenopathy. There is a 1-cm tender oral ulcer with no exudates or thrush. The throat is injected. The remainder of the examination shows no abnormalities. Monospot and rapid streptococcal tests are negative. Which of the following is the most appropriate next step in management?
- ☐ A) Measurement of plasma HIV viral load
 - ☐ B) Serology for herpes simplex virus
 - ☐ C) Throat swab culture for influenza
 - ☐ D) Acyclovir therapy
 - ☐ E) Azithromycin therapy

5. An asymptomatic 67-year-old man comes for a routine health maintenance examination. His last visit was 2 years ago, and he has been healthy since then. His temperature is 37°C (98.6°F), pulse is 65/min, respirations are 16/min, and blood pressure is 150/90 mm Hg. Examination shows mild splenomegaly. Laboratory studies show:

Hematocrit	35%
Leukocyte count	20,000/mm ³
Segmented neutrophils	50%
Bands	10%
Eosinophils	5%
Basophils	5%
Lymphocytes	20%
Metamyelocytes	5%
Myelocytes	5%
Platelet count	700,000/mm ³

Which of the following is the most likely diagnosis?

- ☐ A) Chronic lymphocytic leukemia
- ☐ B) Chronic myelogenous leukemia
- ☐ C) Chronic myelomonocytic leukemia
- ☐ D) Plasma cell myeloma
- ☐ E) Reactive leukocytosis

6. A 47-year-old woman with a 5-year history of rheumatoid arthritis comes to the physician for a follow-up examination. She has morning stiffness in her hands but otherwise feels well. Medications are prednisone and methotrexate. Vital signs are within normal limits. Examination shows mild swelling of the metacarpophalangeal and proximal interphalangeal joints of both hands. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hematocrit	32%
Mean corpuscular volume	75 μm^3
Erythrocyte sedimentation rate	50 mm/h
Serum	
Total iron-binding capacity	150 $\mu\text{g/dL}$ (N=250–400)
Ferritin	300 ng/mL
Iron	30 $\mu\text{g/dL}$

Which of the following is the most appropriate next step in management of this patient's anemia?

- ☐ A) Oral iron therapy
- ☐ B) Subcutaneous erythropoietin therapy
- ☐ C) Transfusion of packed red blood cells
- ☐ D) Colonoscopy
- ☐ E) Bone marrow biopsy
- ☐ F) No further evaluation or treatment is indicated at this time

7. Prior to discharge to a skilled nursing care facility, an 80-year-old woman is evaluated in the hospital because of a decreased hemoglobin concentration. She underwent hemicolectomy 10 days ago for gastrointestinal bleeding secondary to diverticulosis. Four units of packed red blood cells were transfused during the procedure. She has hypertension and type 2 diabetes mellitus. Her medications are lisinopril and metformin. Today, her temperature is 37.2°C (99.0°F), pulse is 88/min, respirations are 14/min, and blood pressure is 130/82 mm Hg. Physical examination shows mucosal pallor and conjunctival icterus. Abdominal examination shows a well-healing incision. Baseline hemoglobin concentration on postoperative day 3 was 10.0 g/dL . Hemoglobin concentration today is 7.5 g/dL . Laboratory studies are most likely to show which of the following additional findings?

- | | Direct Antiglobulin
(Coombs) Test | Serum Bilirubin
Concentration |
|--------------------------|--------------------------------------|----------------------------------|
| ☐ A) | – | increased direct |
| ☐ B) | – | increased indirect |
| ☐ C) | – | normal |
| ☐ D) | + | increased direct |
| ☐ E) | + | increased indirect |
| ☐ F) | Normal | normal |

8. A 52-year-old man is brought to the emergency department 1 hour after a 30-minute episode of weakness in the right arm and leg and difficulty speaking. The symptoms resolved spontaneously. His last visit to a physician was 35 years ago. His pulse is 82/min, and blood pressure is 160/96 mm Hg. Cardiac examination shows a diffuse point of maximal impulse and an S₄ gallop. A left femoral bruit is heard. Neurologic examination shows no focal findings. Laboratory studies show:

Hemoglobin	9.7 g/dL
Serum	
Glucose (nonfasting)	156 mg/dL
Creatinine	3.2 mg/dL
Total cholesterol	210 mg/dL

An ECG shows left ventricular hypertrophy. Treatment of which of the following is most likely to have prevented these findings?

- ☐ A) Congestive heart failure
- ☐ B) Hypercholesterolemia
- ☐ C) Hypertension
- ☐ D) Renal insufficiency
- ☐ E) Type 2 diabetes mellitus

9. An otherwise healthy 92-year-old woman is brought to the physician by her family because of a 1-year history of cognitive decline. Her daughter reports that the patient seems to "fish for words" and often repeats questions. The patient's 94-year-old husband has not noticed a problem. Twenty years ago, she sustained a concussion after falling off a horse. She is a high school graduate. Her 90-year-old sister has significant memory problems; her two younger brothers have no memory problems. The patient takes no medications. On examination, she is pleasant and cooperative. She knows the year, but not the month or date. She can recall a recent holiday. She has no difficulty naming objects by sight, but when asked to name as many animals as she can in 1 minute, she can only name five. She recalls one of three items after 5 minutes. Which of the following is the strongest predisposing factor for this patient's condition?

- ☐ A) Age
- ☐ B) Education level
- ☐ C) Family history
- ☐ D) Gender
- ☐ E) History of head trauma



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

10. A previously healthy 42-year-old man comes to the office because of severe right-sided headaches for the past 3 weeks. He says that he awakens every night at 3 AM with a sharp stabbing pain behind the right eye that usually lasts about 30 minutes. There is no associated nausea. He has smoked two packs of cigarettes daily for 25 years and drinks a six-pack of beer daily. His vital signs are within normal limits. Examination during an episode shows ptosis on the right and tearing and conjunctival injection; the right pupil is 2 mm, and the left is 4 mm. Which of the following is the most likely diagnosis?

- ☐ A) Cluster headache
- ☐ B) Idiopathic intracranial hypertension
- ☐ C) Migraine
- ☐ D) Subdural hematoma
- ☐ E) Temporal lobe glioma
- ☐ F) Tension-type headache



11. A 58-year-old man comes to the office because of a 1-year history of a slowly enlarging, painless, non-itchy lesion on his right inner thigh. He notices the lesion only when he bathes and believes he might be developing a similar one on his right forearm. He has HIV infection well controlled with antiretroviral therapy. He also takes a daily multivitamin. He does not smoke cigarettes or drink alcohol. He used illicit intravenous heroin for 5 years but stopped 35 years ago. He has worked on a farm for 40 years. Vital signs are within normal limits. A photograph of the lesion over the right thigh is shown. Examination of the right forearm shows a 3-cm, raised lesion with well-demarcated borders and dark pigmentation. The remainder of the examination shows no abnormalities. Which of the following historical factors most likely increased this patient's risk for developing these lesions?

- ☐ A) Age
- ☐ B) Gender
- ☐ C) Medical history
- ☐ D) Medication history



11. A 58-year-old man comes to the office because of a 1-year history of a slowly enlarging, painless, non-itchy lesion on his right inner thigh. He notices the lesion only when he bathes and believes he might be developing a similar one on his right forearm. He has HIV infection well controlled with antiretroviral therapy. He also takes a daily multivitamin. He does not smoke cigarettes or drink alcohol. He used illicit intravenous heroin for 5 years but stopped 35 years ago. He has worked on a farm for 40 years. Vital signs are within normal limits. A photograph of the lesion over the right thigh is shown. Examination of the right forearm shows a 3-cm, raised lesion with well-demarcated borders and dark pigmentation. The remainder of the examination shows no abnormalities. Which of the following historical factors most likely increased this patient's risk for developing these lesions?

- ☐ A) Age
- ☐ B) Gender
- ☐ C) Medical history
- ☐ D) Medication history
- ☐ E) Occupation

12. A 67-year-old woman comes to the physician because of persistent actinic keratosis despite treatment with 5-fluorouracil. Examination shows a 0.5 x 0.5-cm lesion. A photograph of the lesion is shown. Which of the following is the most appropriate next step in management?

- ☐ A) Observation only
- ☐ B) Excisional biopsy
- ☐ C) Punch biopsy
- ☐ D) Radiation therapy
- ☐ E) Systemic chemotherapy followed by surgical excision



13. A 47-year-old woman with a 9-year history of rheumatoid arthritis comes to the physician because of a 10-day history of increasing right knee pain and fatigue. She now rates the pain as a 10 on a 10-point scale. She has not had sore throat, cough, shaking chills, or rash. Her temperature is 39.9°C (103.8°F), pulse is 82/min, and blood pressure is 130/80 mm Hg. The right knee is warm to touch with a tense effusion. Arthrocentesis is performed; analysis shows turbid fluid with a poor mucin clot and a leukocyte count of 75,000/mm³ (85% segmented neutrophils). Which of the following is the most likely causal organism?

- ☐ A) *Neisseria gonorrhoeae*
- ☐ B) *Pseudomonas aeruginosa*
- ☐ C) *Staphylococcus aureus*
- ☐ D) *Streptococcus agalactiae* (group B)
- ☐ E) *Yersinia enterocolitica*



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help

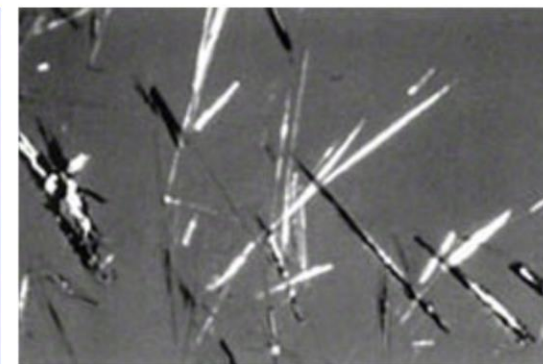


Pause

14. A 42-year-old man comes to the physician because of right shoulder pain for 24 hours. The pain began 8 hours after he spent the day painting his house. The pain increases when he raises his right arm; he is unable to actively abduct the shoulder beyond 60 degrees. Passive range of motion of the shoulder is full and causes only mild discomfort. An x-ray of the shoulder is shown. Which of the following is the most likely diagnosis?

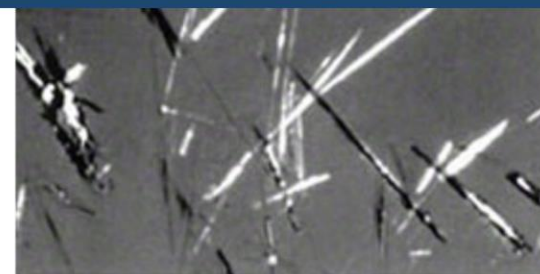
- ☐ A) Biceps tendon rupture
- ☐ B) Cervical nerve root compression
- ☐ C) Glenohumeral subluxation
- ☐ D) Rotator cuff tendinitis
- ☐ E) Torn glenoid labrum





15. A 68-year-old woman is evaluated in the hospital because of right foot pain 1 day after laparoscopic cholecystectomy. She says the pain is sharp and intense, and she has never felt anything like this before. She has no history of trauma to the area. Her temperature is 37.0°C (98.6°F), pulse is 100/min, respirations are 15/min, and blood pressure is 140/70 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. Physical examination shows Heberden nodes of the fingers and crepitus of the knees. The right foot is erythematous, extremely tender to palpation, and warm to touch. Joint aspiration is performed. A photograph of the feet and photomicrograph of the synovial fluid are shown. Administration of which of the following is the most appropriate next step in management?

- ☐ A) Acetaminophen
☐ B) Allopurinol
☐ C) Colchicine



15. A 68-year-old woman is evaluated in the hospital because of right foot pain 1 day after laparoscopic cholecystectomy. She says the pain is sharp and intense, and she has never felt anything like this before. She has no history of trauma to the area. Her temperature is 37.0°C (98.6°F), pulse is 100/min, respirations are 15/min, and blood pressure is 140/70 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. Physical examination shows Heberden nodes of the fingers and crepitus of the knees. The right foot is erythematous, extremely tender to palpation, and warm to touch. Joint aspiration is performed. A photograph of the feet and photomicrograph of the synovial fluid are shown. Administration of which of the following is the most appropriate next step in management?

- ☐ A) Acetaminophen
- ☐ B) Allopurinol
- ☐ C) Colchicine
- ☐ D) Hydroxychloroquine
- ☐ E) Methotrexate

16. An asymptomatic 47-year-old woman comes to the physician for a routine examination. She has hypertension and hypothyroidism. During childhood, she was told she had a heart murmur. Current medications include lisinopril, hydrochlorothiazide, and levothyroxine. She does not smoke cigarettes or drink alcohol. She is 168 cm (5 ft 6 in) tall and weighs 62 kg (136 lb); BMI is 22 kg/m². Her pulse is 70/min, and blood pressure is 120/70 mm Hg. Oral examination shows a dark, cracked, right lower molar with swelling, erythema, and tenderness of the gum. The lungs are clear to auscultation. Cardiac examination shows a normal S₁ and S₂ with variation of S₂ on respiration. The patient is scheduled to undergo a root canal at the dentist next week. Which of the following is the most appropriate recommendation for antibiotic prophylaxis prior to the scheduled dental procedure?
- ☐ A) Oral amoxicillin 1 hour before the dental procedure
 - ☐ B) Oral amoxicillin 24 hours before the procedure
 - ☐ C) Intravenous vancomycin 1 hour before the dental procedure
 - ☐ D) Intravenous vancomycin 24 hours before the dental procedure
 - ☐ E) No antibiotic prophylaxis is indicated

17. A 48-year-old man has a 6-month history of pain in both buttocks when he walks and erectile dysfunction. He has been a heavy smoker for 20 years. His blood pressure in the lower extremities is less than half of that in his arms. His serum cholesterol concentration is 286 mg/dL. Which of the following is the most likely diagnosis?

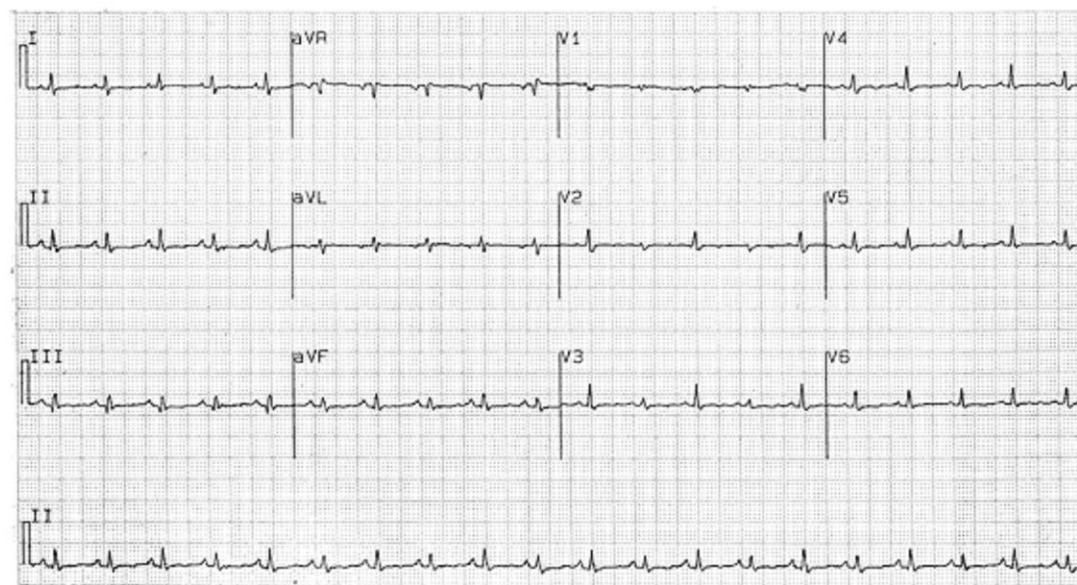
- ☐ A) Aortic aneurysm
- ☐ B) Aortic dissection
- ☐ C) Atherosclerotic narrowing of the distal aorta
- ☐ D) Coarctation of the aorta
- ☐ E) Thromboangiitis obliterans (Buerger disease)

18. A 62-year-old man comes to the emergency department because of a 3-day history of progressive shortness of breath. He has not had chest pain, orthopnea, or paroxysmal nocturnal dyspnea. He completed chemotherapy for small cell carcinoma of the lung 10 months ago. He has a long-term history of nocturia twice nightly that resolved over the past 3 days. He smoked two packs of cigarettes daily for 30 years but quit 1 year ago. His pulse is 116/min, and blood pressure is 96/60 mm Hg. There is jugular venous distention to the angle of the jaw. The lungs are clear to auscultation. Heart sounds are distant. The liver is tender and has a span of 12 cm. There is no pedal edema. Laboratory studies show:

Hemoglobin	10 g/dL
Serum	
Na ⁺	135 mEq/L
Cl ⁻	110 mEq/L
K ⁺	4.2 mEq/L
HCO ₃ ⁻	22 mEq/L
Urea nitrogen	40 mg/dL
Creatinine	1.6 mg/dL

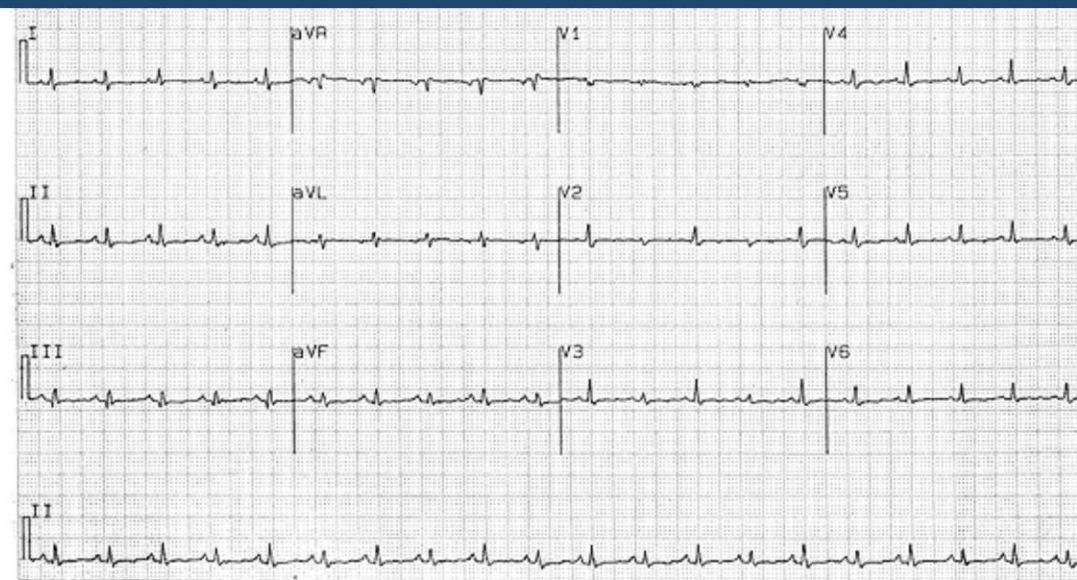
An ECG shows decreased QRS voltage. An x-ray of the chest shows an enlarged cardiac silhouette. The most likely cause of these findings is an abnormality of which of the following?

- ☐ A) Aorta
- ☐ B) Endocardium
- ☐ C) Myocardium
- ☐ D) Pericardium
- ☐ E) Pulmonary artery
- ☐ F) Pulmonary veins



19. A 52-year-old woman comes to the emergency department because of a 1-day history of progressive shortness of breath and right upper abdominal pain. She has been unable to sleep while supine because of difficulty breathing. She underwent chemotherapy 1 year ago for breast cancer. She has chronic obstructive pulmonary disease. She has smoked two packs of cigarettes daily for 30 years. Her temperature is 37.2°C (99°F), pulse is 110/min, respirations are 24/min, and blood pressure is 104/80 mm Hg. Examination shows jugular venous pulsations at the angle of the jaw with the patient sitting upright. The lungs are clear to auscultation. Heart sounds are distant. A grade 2/6 systolic murmur is heard. Abdominal examination shows diffuse hepatomegaly. There is 2+ pedal edema. Her serum urea nitrogen concentration is 42 mg/dL, and serum creatinine concentration is 1.6 mg/dL. An ECG is shown. Which of the following is the most likely explanation for these findings?

- ☐ A) Anterior myocardial infarct
- ☐ B) Cor pulmonale
- ☐ C) Pericardial effusion
- ☐ D) Pulmonary emboli
- ☐ E) Right ventricular myocardial infarct



19. A 52-year-old woman comes to the emergency department because of a 1-day history of progressive shortness of breath and right upper abdominal pain. She has been unable to sleep while supine because of difficulty breathing. She underwent chemotherapy 1 year ago for breast cancer. She has chronic obstructive pulmonary disease. She has smoked two packs of cigarettes daily for 30 years. Her temperature is 37.2°C (99°F), pulse is 110/min, respirations are 24/min, and blood pressure is 104/80 mm Hg. Examination shows jugular venous pulsations at the angle of the jaw with the patient sitting upright. The lungs are clear to auscultation. Heart sounds are distant. A grade 2/6 systolic murmur is heard. Abdominal examination shows diffuse hepatomegaly. There is 2+ pedal edema. Her serum urea nitrogen concentration is 42 mg/dL, and serum creatinine concentration is 1.6 mg/dL. An ECG is shown. Which of the following is the most likely explanation for these findings?

- ☐ A) Anterior myocardial infarct
- ☐ B) Cor pulmonale
- ☐ C) Pericardial effusion
- ☐ D) Pulmonary emboli
- ☐ E) Right ventricular myocardial infarct

20. A previously healthy 42-year-old man is brought to the emergency department because of a 2-hour history of severe abdominal and back pain. His temperature is 37.8°C (100°F), pulse is 120/min and regular, respirations are 22/min, and blood pressure is 160/105 mm Hg in the right arm and 80/50 mm Hg in the left arm. Breath sounds are normal. Cardiac examination shows a normal S₁ and S₂. Femoral pulses are normal on the right and absent on the left. Laboratory studies show:

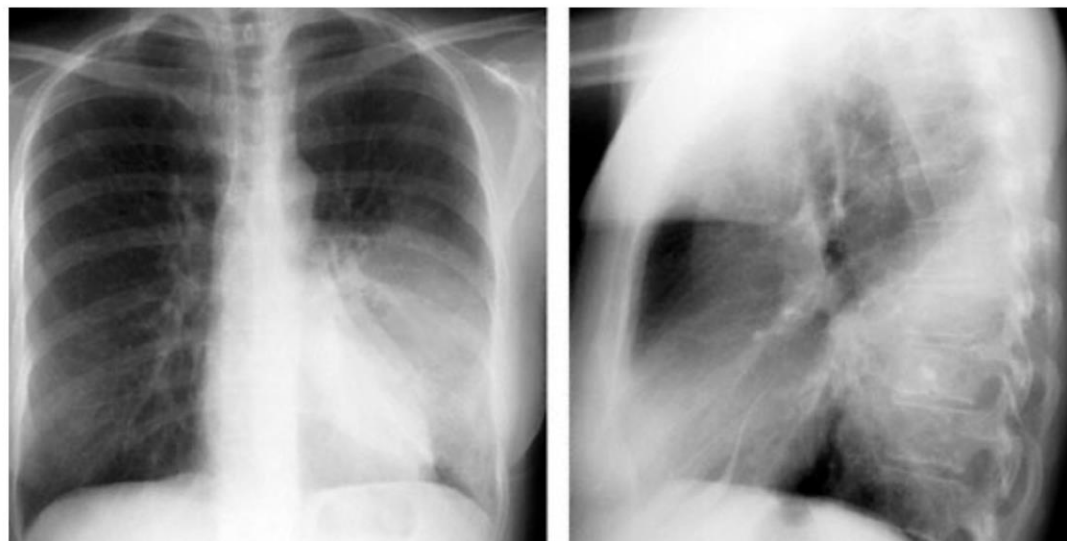
Hematocrit	39%
Leukocyte count	10,500/mm ³
Serum	
Urea nitrogen	32 mg/dL
Creatinine	1.5 mg/dL

Liver function tests are within the reference range. An x-ray of the chest shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Iliac occlusive disease (Leriche syndrome)
- ☐ B) Renal artery embolus
- ☐ C) Ruptured abdominal aortic aneurysm
- ☐ D) Ruptured thoracic aneurysm
- ☐ E) Thoracic aortic dissection

21. A 67-year-old woman is admitted to the hospital 2 hours after the acute onset of pain in the left lower extremity. She has mild mitral insufficiency and chronic atrial fibrillation. She takes no medications. Examination shows a cool left lower extremity and mottling of the skin to the knee. Sensation is absent below the mid thigh. She is unable to move her left foot. After undergoing left femoral embolectomy, sensorimotor function is intact, and left femoral, popliteal, dorsalis pedis, and posterior tibial pulses are palpable. Transesophageal echocardiography shows a dilated left atrium without a patent foramen ovale or atrioventricular thrombus. Which of the following is the most appropriate next step to prevent further embolization in this patient?

- ☐ A) Aspirin therapy
- ☐ B) Digoxin therapy
- ☐ C) Verapamil therapy
- ☐ D) Warfarin therapy
- ☐ E) Mitral valve replacement



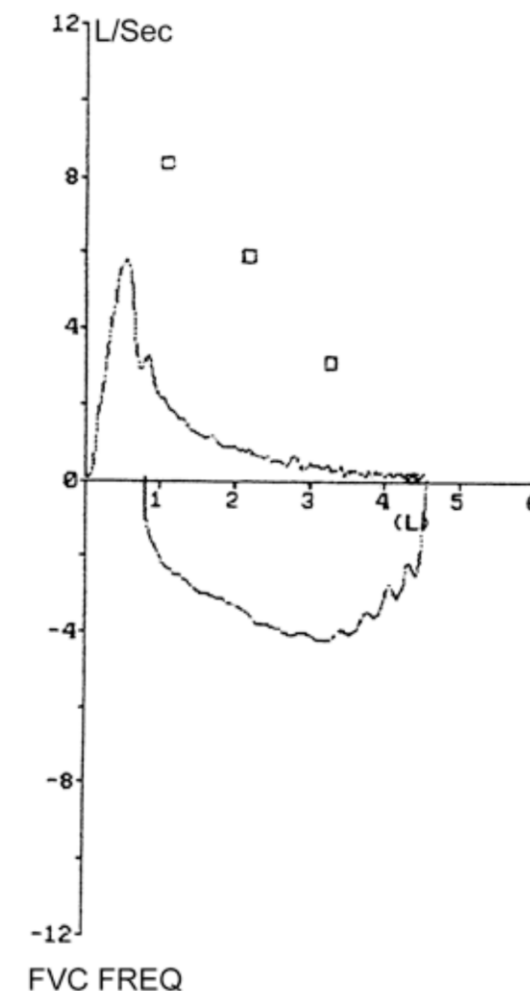
22. One week after undergoing appendectomy and peritoneal lavage for ruptured appendicitis, a previously healthy 61-year-old man is evaluated because of fever, chills, cough, and mild shortness of breath. During the procedure, a peritoneal drain was placed. Postoperatively, intravenous ciprofloxacin and metronidazole therapy was begun. A persistent ileus was diagnosed on postoperative day 2, and a nasogastric tube was inserted. Yesterday, the nasogastric tube was removed, and he was advanced to a regular diet. Today, his temperature is 38.1°C (100.6°F), pulse is 110/min, respirations are 22/min, and blood pressure is 110/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Crackles are heard at the mid left lung field. Abdominal examination shows a clean, dry surgical incision. The peritoneal drain is mostly clear of fluid. Leukocyte count is 15,100/mm³ (65% segmented neutrophils, 10% bands, 25% lymphocytes). Chest x-rays are shown. Which of the following is the most appropriate next step in management?

- ☐ A) Administration of ceftriaxone and azithromycin
- ☐ B) Administration of sulfamethoxazole and cefuroxime
- ☐ C) Administration of vancomycin and ceftazidime
- ☐ D) Bronchoscopy for microbiologic assessment
- ☐ E) CT scan of the chest with pulmonary embolism protocol

23. A 67-year-old man comes to the physician because of nonproductive cough, hemoptysis, and a 9-kg (20-lb) weight loss over the past month. He has smoked two packs of cigarettes daily for 50 years. Breath sounds are decreased over the right posterior hemithorax. Laboratory studies show no abnormalities. An x-ray of the chest shows collapse of the right lower lobe of the lung. Which of the following is the most likely pathophysiology of this patient's condition?
- ☐ A) Endobronchial obstruction
 - ☐ B) Increased pulmonary capillary wedge pressure
 - ☐ C) Pleural inflammation with fluid formation
 - ☐ D) Proximal airway inflammation
 - ☐ E) Pulmonary parenchymal necrosis
 - ☐ F) Pulmonary vascular obstruction

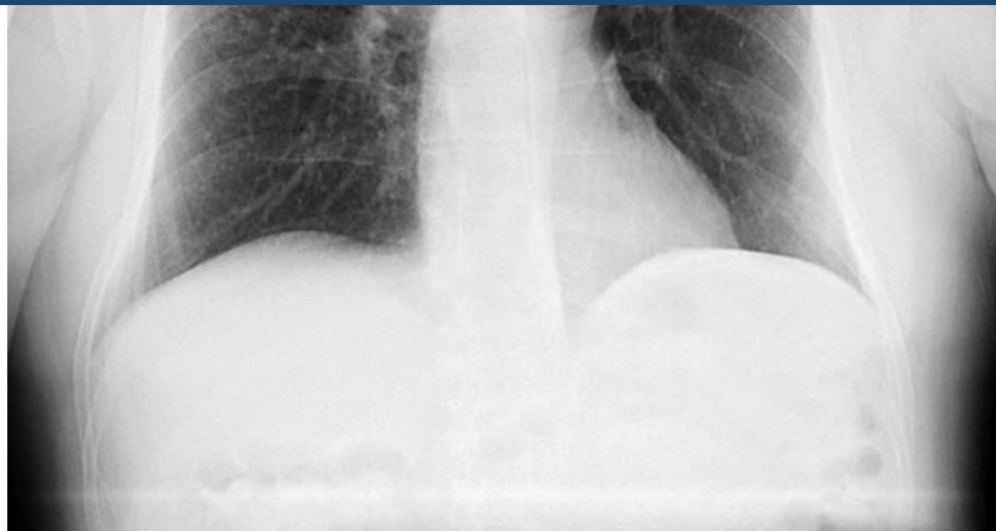
24. A previously healthy 32-year-old man comes to the physician because of a 6-month history of shortness of breath and dry cough with exercise. He has smoked one pack of cigarettes weekly for 5 years. Vital signs are within normal limits. Faint bilateral expiratory wheezes are heard. Pulmonary function tests show a decreased FEV_1 , a normal FVC, and a decreased $FEV_1:FVC$ ratio (45% of predicted). The flow volume loop is shown. Which of the following is the most likely diagnosis?

- ☐ A) Ankylosing spondylitis
- ☐ B) Asthma
- ☐ C) Cystic fibrosis
- ☐ D) Emphysema
- ☐ E) Extrathoracic airway obstruction
- ☐ F) Idiopathic pulmonary fibrosis





25. A 70-year-old man comes to the office because of a 6-month history of progressive shortness of breath with exertion. Six months ago, he was able to walk five blocks before having shortness of breath. He now has shortness of breath after walking one block. He has not had chest pain, palpitations, swelling of his legs, or shortness of breath at night. He has hypertension. His medications are lisinopril and aspirin. He does not smoke cigarettes. He is a retired foundry worker. He is 183 cm (6 ft) tall and weighs 82 kg (180 lb); BMI is 24 kg/m². His pulse is 88/min, respirations are 18/min, and blood pressure is 130/85 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Diffuse crackles are heard bilaterally. Cardiac examination shows no other abnormalities. There is no edema of the extremities. Results of pulmonary function testing show an FVC of 65% of predicted and an FEV₁:FVC ratio of 70%. A chest x-ray is shown. Which of the following is the most likely diagnosis?



25. A 70-year-old man comes to the office because of a 6-month history of progressive shortness of breath with exertion. Six months ago, he was able to walk five blocks before having shortness of breath. He now has shortness of breath after walking one block. He has not had chest pain, palpitations, swelling of his legs, or shortness of breath at night. He has hypertension. His medications are lisinopril and aspirin. He does not smoke cigarettes. He is a retired foundry worker. He is 183 cm (6 ft) tall and weighs 82 kg (180 lb); BMI is 24 kg/m². His pulse is 88/min, respirations are 18/min, and blood pressure is 130/85 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Diffuse crackles are heard bilaterally. Cardiac examination shows no other abnormalities. There is no edema of the extremities. Results of pulmonary function testing show an FVC of 65% of predicted and an FEV₁:FVC ratio of 70%. A chest x-ray is shown. Which of the following is the most likely diagnosis?

- ☐ A) Asbestosis
- ☐ B) Aspergillosis
- ☐ C) Asthma
- ☐ D) Bronchiectasis
- ☐ E) Chronic bronchitis
- ☐ F) Silicosis

26. A 68-year-old woman is brought to the emergency department because of progressively severe cough, fever, and lethargy over the past 5 days. She has a 30-year history of type 2 diabetes mellitus and a 20-year history of rheumatoid arthritis. Her temperature is 38.4°C (101.1°F), pulse is 120/min, respirations are 28/min, and blood pressure is 128/78 mm Hg. She is lethargic. Crackles are heard at the left lung base, and there is dullness to percussion. An x-ray of the chest shows a left lower lobe infiltrate with an adjacent effusion. Pleural fluid analysis shows:

Leukocyte count	50,000/mm ³
Segmented neutrophils	90%
Lymphocytes	10%
Glucose	50 mg/dL
pH	6.99
Lactate dehydrogenase	212 U/L

Which of the following is the most likely underlying cause of these findings?

- ☐ A) Empyema
- ☐ B) Malignancy
- ☐ C) Rheumatoid arthritis
- ☐ D) Tuberculosis
- ☐ E) Viral infection

27. A previously healthy 37-year-old woman comes to the physician because of a 4-month history of abdominal cramps and diarrhea. During this period, she has had a 9-kg (20-lb) weight loss and 8 to 10 loose stools daily that contain a small amount of mucus. Abdominal examination shows mild tenderness of the right lower quadrant. A Sudan black B fat stain of the stool is positive. Colonoscopy with biopsy of the terminal ileum shows inflammation with granuloma formation. Which of the following is the most likely mechanism of this patient's steatorrhea?

- ☐ A) Bacterial overgrowth
- ☐ B) Coexisting pancreatic exocrine dysfunction
- ☐ C) Defective release of secretin
- ☐ D) Ineffective reabsorption of bile salts
- ☐ E) Poor mixing of bile salts with dietary lipids

28. A 37-year-old man comes to the physician because of a 1-week history of progressive left-sided facial pain and swelling. Initially, the pain was only precipitated by eating, but now it has been constant for the past 3 days. He has no history of serious illness and takes no medications. His temperature is 37.3°C (99.2°F), pulse is 88/min, respirations are 16/min, and blood pressure is 136/80 mm Hg. Examination shows an area of edema over the lateral left face that is warm, tender, and erythematous. Which of the following is the most likely diagnosis?

- ☐ A) Mumps
- ☐ B) Sarcoidosis
- ☐ C) Sialadenitis
- ☐ D) Sjögren syndrome
- ☐ E) Temporomandibular joint dysfunction

29. An obese 42-year-old woman has an episode of colicky pain in the right upper quadrant of the abdomen that lasts 2 hours and subsides spontaneously. She has no fever or leukocytosis and no signs of peritoneal irritation in the right upper quadrant. Liver function tests show no abnormalities. Ultrasonography shows gallstones. Which of the following is the most likely cause of the abdominal pain?

- ☐ A) Bacterial growth in bile within the common hepatic duct
- ☐ B) Bacterial invasion of the gallbladder wall
- ☐ C) Transitory chemical irritation of the gallbladder wall
- ☐ D) Transitory obstruction of the common hepatic duct by gallstones
- ☐ E) Transitory obstruction of the cystic duct by gallstones

30. A 52-year-old man comes to the physician because of a 4-month history of loose, foul-smelling stools; during this period, he has had a 6.3-kg (14-lb) weight loss despite a good appetite. He also has had urinary frequency and urgency for 3 weeks. He has not had fever, chills, nausea, or vomiting. He has a 10-year history of recurrent alcoholic pancreatitis that required hospitalization once or twice each year because of severe abdominal pain. He has not been hospitalized since he abstained from drinking alcohol 2 years ago, but he has had chronic mild midepigastic pain since that time. During his last hospitalization 2 years ago, an x-ray of the abdomen showed calcifications in the pancreas. He is 175 cm (5 ft 9 in) tall and weighs 69 kg (153 lb); BMI is 23 kg/m². Examination shows temporal wasting. Cardiopulmonary examination shows no abnormalities. Abdominal examination shows midepigastic tenderness with no masses or hepatosplenomegaly; bowel sounds are normal. There is no peripheral edema. Laboratory studies show:

Hematocrit	39%
Mean corpuscular volume	84 μm^3
Leukocyte count	9000/mm ³
Segmented neutrophils	72%
Lymphocytes	18%
Monocytes	10%
Platelet count	230,000/mm ³
Serum	
Urea nitrogen	17 mg/dL
Glucose	280 mg/dL
Creatinine	1.2 mg/dL
Protein, total	5.8 g/dL
Albumin	2.7 g/dL

Which of the following sets of findings is the most likely explanation for this patient's increased serum glucose concentration?

- | | Insulin
Production | Glucagon
Production | Peripheral Insulin
Response |
|--------------------------|-------------------------------|--------------------------------|--|
| ☐ A) | Decreased | decreased | normal |
| ☐ B) | Decreased | increased | normal |
| ☐ C) | Increased | decreased | decreased |
| ☐ D) | Increased | increased | decreased |



2 years ago, but he has had chronic mild midepigastric pain since that time. During his last hospitalization 2 years ago, an x-ray of the abdomen showed calcifications in the pancreas. He is 175 cm (5 ft 9 in) tall and weighs 69 kg (153 lb); BMI is 23 kg/m². Examination shows temporal wasting. Cardiopulmonary examination shows no abnormalities. Abdominal examination shows midepigastric tenderness with no masses or hepatosplenomegaly; bowel sounds are normal. There is no peripheral edema. Laboratory studies show:

Hematocrit	39%
Mean corpuscular volume	84 µm ³
Leukocyte count	9000/mm ³
Segmented neutrophils	72%
Lymphocytes	18%
Monocytes	10%
Platelet count	230,000/mm ³
Serum	
Urea nitrogen	17 mg/dL
Glucose	280 mg/dL
Creatinine	1.2 mg/dL
Protein, total	5.8 g/dL
Albumin	2.7 g/dL

Which of the following sets of findings is the most likely explanation for this patient's increased serum glucose concentration?

- | | Insulin
Production | Glucagon
Production | Peripheral Insulin
Response |
|--------------------------|-----------------------|------------------------|--------------------------------|
| ☐ A) | Decreased | decreased | normal |
| ☐ B) | Decreased | increased | normal |
| ☐ C) | Increased | decreased | decreased |
| ☐ D) | Increased | increased | decreased |
| ☐ E) | Increased | normal | decreased |
| ☐ F) | Normal | normal | decreased |

31. A 23-year-old woman comes to the office because of a 6-week history of fatigue and swelling of her eyelids, ankles, and feet. She has no history of serious illness and takes no medications. She is 163 cm (5 ft 4 in) tall and weighs 54 kg (120 lb); BMI is 21 kg/m². Vital signs are within normal limits. Examination shows 2+ periorbital edema and 3+ pedal edema. No other abnormalities are noted. Fasting serum studies show a total cholesterol concentration of 348 mg/dL and triglyceride concentration of greater than 400 mg/dL. This patient is at greatest risk for which of the following complications?

- ☐ A) Deep venous thrombosis
- ☐ B) Hypothyroidism
- ☐ C) Nephrolithiasis
- ☐ D) Peripheral vascular disease
- ☐ E) Stasis dermatitis

32. A 20-year-old man is brought to the emergency department unconscious 30 minutes after a motor vehicle collision. His temperature is 36.2°C (97.2°F), pulse is 150/min, respirations are 18/min, and palpable systolic blood pressure is 60 mm Hg. Examination shows multiple lacerations and contusions over the trunk and upper and lower extremities and abdominal distention. After administration of intravenous fluids, splenectomy, and transfusion of 4 units of packed red blood cells, his condition stabilizes. Twenty-four hours later, his pulse is 88/min, and blood pressure is 120/70 mm Hg. Abdominal examination shows a normal-appearing surgical incision. Over the past 12 hours, his urine output has been 100 mL. Laboratory studies show:

Serum	
Na ⁺	140 mEq/L
K ⁺	4.2 mEq/L
Urea nitrogen	30 mg/dL
Creatinine	2.2 mg/dL
Urine	
Na ⁺	70 mEq/L
Osmolality	280 mOsmol/kg

Which of the following is the most likely diagnosis?

- ☐ A) Acute tubular necrosis
- ☐ B) Hypovolemia
- ☐ C) Interstitial nephritis
- ☐ D) Transfusion reaction
- ☐ E) Urinary tract obstruction

33. A 77-year-old man with osteoarthritis comes to the office because of a 2-week history of increasing right knee pain. He was diagnosed with osteoarthritis 5 years ago. Last year, an x-ray of the right knee showed mild degenerative changes. He has been treated with increasing doses of ibuprofen and now takes 800 mg of ibuprofen three times daily. He tries to avoid walking because of the pain. He has not had abdominal pain, change in appetite, or any other symptoms. His temperature is 37.0°C (98.6°F). On examination, flexion of the right knee is limited to 20 degrees. There are no effusions. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hemoglobin	13.6 g/dL
Leukocyte count	8500/mm ³
Serum	
Urea nitrogen	32 mg/dL
Creatinine	3 mg/dL
Urine	
Blood	negative
Protein	negative
RBC	none
WBC	none

Which of the following is the most appropriate next step in management?

- ☐ A) Continue ibuprofen at the current dose and repeat the laboratory studies in 2 weeks
- ☐ B) Continue ibuprofen at the current dose and begin oral hydrocortisone
- ☐ C) Decrease the dose of ibuprofen and begin oral prednisone
- ☐ D) Discontinue ibuprofen and administer an intra-articular injection of triamcinolone
- ☐ E) Switch from ibuprofen to oral celecoxib

34. An 82-year-old man comes to the physician because of a 1-month history of urinary frequency and dribbling of small amounts of urine. He has not had pain on urination or fever. He is alert and oriented. His temperature is 36.7°C (98°F), pulse is 86/min, respirations are 16/min, and blood pressure is 145/90 mm Hg. Rectal examination shows an enlarged, smooth, nontender prostate. The remainder of the examination shows no abnormalities. Bladder catheterization shows a postvoid residual volume of 750 mL. Urinalysis is within normal limits. Which of the following is the most likely cause of this patient's incontinence?

- ☐ A) Bladder neck obstruction
- ☐ B) Detrusor hyperactivity
- ☐ C) Neurogenic bladder
- ☐ D) Prostatic infection
- ☐ E) Urethral hypermobility

35. A 67-year-old man comes to the physician because of a 2-day history of fever, nausea, a continuous urge to void, and progressive difficulty with urination. Every 30 minutes, he passes a small amount of urine and has an uncomfortable burning sensation. He says he has slept poorly during this period. Prior to the onset of his current symptoms, he had a 12-week history of difficulty initiating a urine stream and of awakening two to three times nightly to urinate. He has no history of serious illness and takes no medications. His temperature is 38.2°C (100.8°F), pulse is 74/min, and blood pressure is 170/90 mm Hg. Cardiac examination shows no abnormalities. Abdominal examination shows suprapubic fullness. Rectal examination shows a symmetrically enlarged prostate with no tenderness or masses. Laboratory studies show:

Hematocrit	45%
Leukocyte count	18,300/mm ³
Platelet count	230,000/mm ³
Serum	
Na ⁺	132 mEq/L
K ⁺	5.3 mEq/L
Cl ⁻	103 mEq/L
HCO ₃ ⁻	21 mEq/L
Urea nitrogen	78 mg/dL
Glucose	134 mg/dL
Creatinine	4.8 mg/dL
Urine	
Glucose	negative
Protein	1+
RBC	30/hpf
WBC	50/hpf

Which of the following is the most appropriate initial step in management?

- ☐ A) Ultrasonography of the kidneys
- ☐ B) CT scan of the pelvis
- ☐ C) Furosemide therapy
- ☐ D) Levofloxacin therapy
- ☐ E) Placement of a urinary catheter

a 12-week history of difficulty initiating a urine stream and of awakening two to three times nightly to urinate. He has no history of serious illness and takes no medications. His temperature is 38.2°C (100.8°F), pulse is 74/min, and blood pressure is 170/90 mm Hg. Cardiac examination shows no abnormalities. Abdominal examination shows suprapubic fullness. Rectal examination shows a symmetrically enlarged prostate with no tenderness or masses. Laboratory studies show:

Hematocrit	45%
Leukocyte count	18,300/mm ³
Platelet count	230,000/mm ³
Serum	
Na ⁺	132 mEq/L
K ⁺	5.3 mEq/L
Cl ⁻	103 mEq/L
HCO ₃ ⁻	21 mEq/L
Urea nitrogen	78 mg/dL
Glucose	134 mg/dL
Creatinine	4.8 mg/dL
Urine	
Glucose	negative
Protein	1+
RBC	30/hpf
WBC	50/hpf

Which of the following is the most appropriate initial step in management?

- ☐ A) Ultrasonography of the kidneys
- ☐ B) CT scan of the pelvis
- ☐ C) Furosemide therapy
- ☐ D) Levofloxacin therapy
- ☐ E) Placement of a urinary catheter
- ☐ F) Hemodialysis

36. A previously healthy 22-year-old woman comes to the emergency department because of a 7-day history of vaginal discharge and a 3-day history of pustular lesions over her chest and arms. She has been sexually active with a new male partner for 2 weeks; they do not use condoms. Her only medication is an oral contraceptive. On arrival, her temperature is 38.3°C (101°F), pulse is 90/min, and blood pressure is 100/70 mm Hg. Examination shows approximately 10 pustular lesions over the thorax and upper extremities. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Azithromycin
- ☐ B) Ceftriaxone
- ☐ C) Nafcillin
- ☐ D) Penicillin
- ☐ E) Vancomycin

37. A 28-year-old woman comes to the office because of an 8-week history of fatigue and cold intolerance. She has a 2-year history of bipolar disorder. At the age of 4 years, she was hospitalized for 2 days for treatment of meningitis. At the age of 16 years, she had loss of consciousness after a motor vehicle collision. Her medications are divalproex, an oral contraceptive, and a daily multivitamin. Her brother has type 1 diabetes mellitus. The patient does not smoke cigarettes. She drinks two 12-oz beers daily and smokes marijuana four times monthly. She appears well developed and well nourished. Her temperature is 36.4°C (97.5°F), pulse is 54/min, respirations are 14/min, and blood pressure is 120/74 mm Hg. The thyroid gland is nontender and cannot be palpated. Deep tendon reflexes are 2+. The remainder of the examination, including cardiac examination, shows no abnormalities. Serum studies show a thyroid-stimulating hormone concentration of 23 $\mu\text{U/mL}$ and free thyroxine concentration of 0.5 ng/dL (N=0.8–2.4). Which of the following historical factors has most likely increased this patient's risk for her current condition?

- ☐ A) Alcohol consumption
- ☐ B) Childhood illness
- ☐ C) Current medications
- ☐ D) Family history
- ☐ E) History of motor vehicle collision
- ☐ F) Illicit drug use



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

38. A 42-year-old woman is admitted to the hospital because of a 1-month history of burning and tingling of the skin over all the extremities and fatigue. She has hypertension and chronic low back pain. Her only medication is hydrochlorothiazide. She appears tired but is not in acute distress. Her temperature is 37.2°C (99°F), pulse is 88/min, respirations are 14/min, and blood pressure is 110/80 mm Hg. Tapping over the facial nerve produces an involuntary contraction of the ipsilateral muscles. The remainder of the examination shows no abnormalities. Serum studies show:

Na ⁺	138 mEq/L
K ⁺	4.1 mEq/L
Cl ⁻	100 mEq/L
Ca ²⁺	7.1 mg/dL
Phosphate	6 mg/dL
Protein, total	9 g/dL
Albumin	5 g/dL

Which of the following is the most likely explanation for this patient's condition?

- ☐ A) Autonomous secretion of aldosterone
- ☐ B) Autonomous secretion of parathyroid hormone
- ☐ C) Deficiency of parathyroid hormone
- ☐ D) Deficiency of vitamin B₁₂ (cobalamin)
- ☐ E) Excess secretion of 25-hydroxyvitamin D
- ☐ F) Excess secretion of magnesium

39. A 32-year-old woman, gravida 1, para 1, comes to the physician because she has not been able to conceive for 12 months. Her son was born 4 years ago. Her menses were normal until 1 year ago when they became irregular. She has not had a menstrual period for the past 6 months. She has also had a small amount of milky secretion from her breasts. She does not take medications. Her vital signs are within normal limits. Examination shows no abnormalities except for galactorrhea. Her serum prolactin concentration is 200 ng/mL. A pregnancy test is negative. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Cytologic examination of breast secretions
- ☐ B) Measurement of serum estradiol concentration
- ☐ C) Measurement of serum follicle-stimulating and luteinizing hormone concentrations
- ☐ D) Mammography
- ☐ E) MRI of the pituitary gland

40. A 32-year-old woman comes to the emergency department 12 hours after the sudden onset of fever, diffuse muscle and bone pain, and dizziness. She has no history of serious illness and takes no medications. Her temperature is 39°C (102.2°F), pulse is 100/min, and blood pressure is 90/50 mm Hg while supine. Her palpable systolic blood pressure is 70 mm Hg while sitting. Examination shows a diffuse, erythematous, macular rash over the face, trunk, hands, and feet. There is desquamation over the palms of the hands. The lungs are clear to auscultation. Cardiac examination shows no murmurs or gallops. There is 1+ peripheral edema bilaterally. Pelvic and neurologic examinations show no abnormalities. Which of the following is the most appropriate next step in treatment?

- ☐ A) Oral prednisone
- ☐ B) Intravenous dopamine
- ☐ C) Intravenous fluids
- ☐ D) Intravenous metronidazole
- ☐ E) Intravenous penicillin

41. An 82-year-old woman comes to the office because of a 3-week history of bilateral frontal headaches and a 3-day history of pain over the parietal skull regions. She also has claudication of the jaw and tongue after she chews vigorously. Examination and complete blood count 6 months ago showed no abnormalities. Today, her temperature is 38.5°C (101.3°F), pulse is 90/min and regular, respirations are 20/min, and blood pressure is 160/85 mm Hg. On physical examination, there is tenderness of the temporal regions bilaterally with the left more tender than the right. Results of laboratory studies show:

Hematocrit	27%
Mean corpuscular volume	87 μm^3
Leukocyte count	5800/ mm^3 with a normal differential
Platelet count	450,000/ mm^3
Erythrocyte sedimentation rate	78 mm/h

Which of the following is the most likely ophthalmic complication of this disorder?

- ☐ A) Dendritic corneal ulceration
- ☐ B) Hyperviscosity of the retinal veins
- ☐ C) Ischemic optic neuropathy
- ☐ D) Scleromalacia perforans
- ☐ E) Uveitis

42. A 27-year-old woman comes to the physician because of a 6-month history of her hands and fingers becoming tender and turning white on exposure to cold. The symptoms occur four times monthly. She has not had any other symptoms. She has no history of serious illness and takes no medications. She has smoked one pack of cigarettes daily for 8 years. Examination shows no abnormalities. Results of laboratory studies are within the reference range. In addition to smoking cessation, which of the following is the most appropriate recommendation?

- ☐ A) Dressing warmly in cold weather
- ☐ B) Doxazosin therapy
- ☐ C) Prednisone therapy
- ☐ D) Propranolol therapy
- ☐ E) Sympathectomy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

43. A 27-year-old man, who works for a lawn maintenance service, is brought to the emergency department by his supervisor 10 minutes after he was found incoherent and wandering around the neighborhood where they were working in the late afternoon. His supervisor reports that the patient was fine when he arrived at work at 6:30 AM, but no one could locate him after they took a lunch break at noon. The temperature outside is 103°F. He takes an over-the-counter antihistamine for severe allergies. His father died of a myocardial infarction at the age of 39 years. His mother and sister have autoimmune thyroiditis. He is alert but not oriented to person, place, or time. His temperature is 39.4°C (103°F), pulse is 100/min, respirations are 18/min, and blood pressure is 120/68 mm Hg. Examination shows warm, dry skin. The pupils are 4 mm and sluggishly reactive. Neurologic examination shows no muscle rigidity. Which of the following is the most likely diagnosis?

- ☐ A) Heat stroke
- ☐ B) Malignant hyperthermia
- ☐ C) Myocardial ischemia
- ☐ D) Neuroleptic malignant syndrome
- ☐ E) Thyroid storm

44. A 24-year-old man who is employed as a landscaper is brought to the emergency department by a coworker 20 minutes after he collapsed while working in a garden; he had light-headedness just before he lost consciousness. On arrival, he is responsive but is unable to give a medical history. His temperature is 37.2°C (99°F), pulse is 125/min, respirations are 28/min, and palpable systolic blood pressure is 70 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Wheezes are heard bilaterally. Which of the following is the most appropriate immediate step in management?

- ☐ A) Arterial blood gas analysis
- ☐ B) Determination of eosinophil count
- ☐ C) Determination of peak expiratory flow rate
- ☐ D) Measurement of plasma histamine concentration
- ☐ E) Toxicology screening
- ☐ F) Administration of corticosteroids
- ☐ G) Administration of diphenhydramine
- ☐ H) Administration of epinephrine

45. A 74-year-old woman is admitted to the hospital following a myocardial infarction. She is confused and has cool, clammy skin. Over the next hour, her blood pressure decreases from 148/74 mm Hg to 80/40 mm Hg. Arterial blood gas analysis is most likely to show which of the following?

- ☐ A) Hypoxemia with normal pH
- ☐ B) Metabolic acidosis
- ☐ C) Metabolic alkalosis
- ☐ D) Respiratory acidosis
- ☐ E) Respiratory alkalosis

46. A 52-year-old man is brought to the emergency department because of a 1-day history of fever and chills. He received the diagnosis of small cell carcinoma of the lung 1 month ago and is undergoing chemotherapy. A right subclavian central venous catheter is in place. His temperature is 38.9°C (102.1°F), pulse is 124/min, respirations are 24/min, and blood pressure is 104/68 mm Hg. Examination shows warm, flushed skin. The catheter site is slightly erythematous, and there is tenderness over the subcutaneous pocket. His leukocyte count is 12,300/mm³. A chest x-ray shows a nodule in the right upper lobe. Which of the following is the most likely causal organism?

- ☐ A) *Candida albicans*
- ☐ B) *Clostridium perfringens*
- ☐ C) *Escherichia coli*
- ☐ D) *Pseudomonas aeruginosa*
- ☐ E) *Staphylococcus epidermidis*



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

47. A 37-year-old woman comes to the physician for HIV screening. She is asymptomatic. She has no history of serious illness and takes no medications. She does not drink alcohol or use illicit drugs. She has had two lifetime male sexual partners, and they used condoms consistently. Examination shows no abnormalities. An ELISA is positive for HIV; the sensitivity of the ELISA is 95%, and the specificity is 90%. Which of the following represents the most appropriate sensitivity and specificity of an additional test to confirm this patient's diagnosis?

- | | Sensitivity | Specificity |
|--------------------------|--------------------|--------------------|
| ☐ A) | 95% | 35% |
| ☐ B) | 99% | 90% |
| ☐ C) | 99% | 95% |
| ☐ D) | 99% | 99% |

48. A study is conducted to assess a possible association between regular use of alcohol and colon cancer. Cases consist of patients hospitalized with colon cancer; controls consist of age- and gender-matched patients admitted to the same hospital for treatment of an acute injury. Cases and controls are interviewed after their conditions stabilize. The researchers conclude that regular use of alcohol reduced the risk of colon cancer. Which of the following is the most important reason to be concerned about the authors' conclusion?

- ☐ A) Confounding by age
- ☐ B) Inappropriate case group
- ☐ C) Inappropriate control group
- ☐ D) Lack of incidence rates for alcohol users and nonusers

49. A 77-year-old man has had intermittent confusion since transfer from the hospital to hospice care 6 days ago. His blood pressure has ranged between 90/45 mm Hg and 70/20 mm Hg during this period. He was treated in the hospital for a myocardial infarction sustained 3 weeks ago. During the past 8 years, he has been admitted to the hospital four times for myocardial infarction. He has been treated unsuccessfully for atrial and ventricular arrhythmias and has been unable to tolerate inotropic agents. He has had left hemiparesis since he sustained a stroke 30 years ago. His physician has explained to him and his family that he is not a candidate for cardiac transplant or implant of a left ventricular assistance device. The patient has severe chronic back pain. His only medication is morphine. The patient's advance directive designates his daughter as his health care proxy, and the patient has explained that he wants his daughter to make medical decisions for him. The daughter says that the hospice care is insufficient in controlling her father's pain and asks that he be admitted to the hospital. Which of the following is the most appropriate next step in management?

- ☐ A) Discuss the patient's pain control with the hospice staff
- ☐ B) Spiritual counseling
- ☐ C) Psychiatric assessment
- ☐ D) Admission to the hospital
- ☐ E) Echocardiography and feeding tube placement

50. A 54-year-old man is hospitalized because of a 5-day history of chills and moderate pain, edema, and erythema of the right foot. His symptoms began 2 days after he stepped on a nail. He has coronary artery disease, type 2 diabetes mellitus, hypertension, and hyperlipidemia. He has no known drug allergies. His medications are metformin, insulin, lisinopril, metoprolol, atorvastatin, and aspirin. His temperature is 38.5°C (101.3°F), pulse is 110/min, respirations are 16/min, and blood pressure is 145/92 mm Hg. Examination of the right lower extremity shows erythema, warmth, and tenderness to palpation from the proximal aspect of the ankle to the dorsum of the foot. The remainder of the examination shows no abnormalities. Intravenous vancomycin and piperacillin-tazobactam therapy is begun, and the patient's home medication regimen is continued. While filling the prescription, the pharmacist notices a problem in the medication order. Investigation shows that the physician wrote the prescription in an immediate-release formulation instead of an extended-release formulation. Which of the following best characterizes this type of medical error?

- ☐ A) Active failure
- ☐ B) Adverse event
- ☐ C) Latent condition
- ☐ D) Near miss
- ☐ E) Sentinel event